NUMBER SYSTEMS

Conversion from\nBinary to Octal

The grouping method | Base 2 → Base 8

LEARN • FOLLOW • PRACTISE

A step-by-step guide with worked examples

Learning goals

By the end of this lesson, you should be able to...

01

Explain

why binary digits are grouped in threes when converting to octal.

02

Group

binary digits correctly from right to left.

03

Convert

each 3-bit group into its matching octal digit.

04

Solve

binary-to-octal conversions and check the answer.

What is the grouping method?

A shortcut for converting binary numbers into octal numbers.

Why groups of 3?

Because $8 = 2^3$, one octal digit represents exactly three binary digits.

000	001	010
= 0	= 1	= 2

Remember

1. Start grouping from the RIGHT.

2. Make groups of THREE bits.

3. Add leading zeros if needed.

4. Convert each group and read left to right.

Binary-to-octal reference

Each 3-bit binary group matches one octal digit.

Binary group	Octal digit	Binary group	Octal digit
000	0	100	4
001	1	101	5
010	2	110	6
011	3	111	7

Tip: You only need to memorise the values from 000 to 111.

Worked example 1: No extra zero needed

Convert $(110101)_2$ to octal.

STEP 1 **Group from the RIGHT in sets of three.**

110 ↓ **$110_2 = 6$**

101 ↓ **$101_2 = 5$**

FINAL ANSWER

$(65)_8$

STEP 2 Convert each group: $110_2 = 6_8$ and $101_2 = 5_8$

Therefore, $(110101)_2 = (65)_8$

Worked example 2: Add a zero on the LEFT

Convert $(10110)_2$ to octal.

STEP 1 **Write the binary number: 10110**

STEP 2 Group from the right. The first group has only two bits.

10 → add 0 **010** **110**

STEP 3 $010_2 = 2_8$ $110_2 = 6_8$

ANSWER

$(26)_8$

Therefore, $(10110)_2 = (26)_8$

Worked example 3: Three octal digits

Convert $(10011101)_2$ to octal.

Step 1 — Group into threes from the right

Binary: 10011101

Groups: 010 011 101

(A zero was added to the LEFT to complete the first group.)

$010_2 = 2_8$

$011_2 = 3_8$

$101_2 = 5_8$

Final answer: $(10011101)_2 = (235)_8$

Let’s practise!

Try each question first, then check the worked answers.

A **Convert $(111000)_2$ to octal**

Groups: 111 | 000

Convert: 7 | 0

$(70)_8$

B **Convert $(1001)_2$ to octal**

Groups: 001 | 001

Convert: 1 | 1

$(11)_8$

C **Convert $(11001110)_2$ to octal**

Groups: 011 | 001 | 110

Convert: 3 | 1 | 6

$(316)_8$

Remember: group from RIGHT → add leading zeros if needed → convert → combine.

QUICK RECAP

Four steps to remember

1. Start at the RIGHT side of the binary number.
2. Separate the digits into groups of THREE.
3. Add zeros to the LEFT of the first group if needed.
4. Convert each group and combine the octal digits.

Exit question

Convert $(111101)_2$ to octal.

Answer: 111 | 101 → 7 | 5 → $(75)_8$

Thank you!